

Eigenvalue & Eigenvectors

(1) (a) True. λ is an eigenvalue of $A \Leftrightarrow \exists v \neq 0$, st. $(A - \lambda I)v = 0$ has
 $\Leftrightarrow \det(A - \lambda I) = 0 \Leftrightarrow \det(A - \lambda I)^T = 0 \Leftrightarrow \det(A^T - \lambda I) = 0$

$\Leftrightarrow \exists w \neq 0$, st. $(A^T - \lambda I)w = 0$. i.e. λ is an eigenvalue of A^T $\square$

(b) False. $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is an eigenvector, since $Av = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = v \neq 0$.

However, $A^T v = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \neq \lambda v$ for some $\lambda \in \mathbb{R}$.

v is not an eigenvector of A^T

$\rightarrow \leftarrow$

(2) (a) It suffices to show the given set is linear independent.

Consider, $a_i \in \mathbb{R}$, with

$$\sum_{i=0}^4 a_i A^i v = 0 \quad \textcircled{1}$$

By apply left multiplying A on $\textcircled{1}$, we obtain

$$a_0 A v + a_1 A^2 v + a_2 A^3 v + a_3 A^4 v + \cancel{a_4 A^5 v} = 0$$

Again, one can deduce

$$a_0 A^4 v = 0,$$

since $A^4 v \neq 0$, we have $a_0 = 0$

By similar argument, $a_0 = a_1 = a_2 = a_3 = a_4 = 0$, i.e.

$\{v, \dots, A^4 v\}$ is a linear independent set, whose span is of dimension 5, Therefore it is a basis of $\mathbb{R}^5$ $\square$

(b)

$$[L_A]_{\beta} = \begin{pmatrix} [L_{A^0 v}]_{\beta} & [L_{A(Av)}]_{\beta} & [L_{A(A^2 v)}]_{\beta} & [L_{A(A^3 v)}]_{\beta} & [L_{A(A^4 v)}]_{\beta} \end{pmatrix}$$

$\nearrow$ see revision class 2

Ex How to compute (say) $[L_A(A^2v)]_\beta$?

$$L_A(A^2v) = A(A^2v) = A^3v = 0 \times v + 0 \times Av + 0 \times A^2v + 1 \times A^3v + 0 \times A^4v$$

$$\Rightarrow [L_A(A^2v)]_\beta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{Therefore, } [L_A]_\beta = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(c) \det([L_A]_\beta - \lambda I) = 0 \Leftrightarrow \lambda = 0 \quad \checkmark$$

Remark = This is so-called "Nilpotent matrix", i.e. a matrix A which vanishes at some power ($\exists n \in \mathbb{N}$ s.t. $A^n = 0$)

The above result shows any nilpotent matrix is similar to a matrix with all entries being zero except for the sub-diagonal being 1, i.e.

$$\begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{pmatrix}_{n \times n}$$

This matrix plays an vital role in proving

"Jordan - Canonical Form"

which shall be taught in Math 2102

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(3). We denote $m_a(\lambda)$ to be the algebraic multiplicity of λ as an eigenvalue of $A \in M_n(\mathbb{R})$

Recall that: $A \in M_n(\mathbb{R})$ is diagonalizable iff $\exists$ basis $\beta = \{v_1, \dots, v_n\}$ s.t. v_i is eigenvector of A

Before we solve the problems, let me give some general results for the completeness of the Lecture note. $\cup$

Remark 1

$$\sum_{\lambda \text{ eigenvalue}} m_a(\lambda) \leq n$$

the equality holds iff the characteristic poly. factorizes a product of linear terms over $\mathbb{R}$.

Thm 1.31 (with proof): Let λ be an eigenvalue of a $n \times n$ real matrix A . then $1 \stackrel{\textcircled{1}}{\leq} \dim E_\lambda \stackrel{\textcircled{2}}{\leq} m_a(\lambda) \stackrel{\textcircled{3}}{\leq} n$

- ①: $\dim E_\lambda = \text{Nullity}(A - \lambda I) \geq 1$ since $\det(A - \lambda I) = 0$
②: Set $\{v_1, \dots, v_m\}$ be the basis of E_λ , by extension thm (Example sheet)

It can be extended to $\beta = \{v_1, \dots, v_m, v_{m+1}, \dots, v_n\}$, a basis of $\mathbb{R}^n$.

Denote $P = \begin{bmatrix} | & | & | & | & | \\ v_1 & v_2 & \dots & v_m & v_{m+1} & \dots & v_n \\ | & | & | & | & | \end{bmatrix}$, P is invertible

$$\text{we have } AP = \begin{bmatrix} Av_1 & Av_2 & Av_3 & \dots & Av_n \end{bmatrix} = \begin{bmatrix} \lambda v_1 & \dots & \lambda v_m & Av_{m+1} & \dots & Av_n \end{bmatrix}$$

$$= P \begin{bmatrix} \lambda I_m & B \\ 0 & C \end{bmatrix}$$

for some block matrix B , and C .

$$\Rightarrow P^{-1}AP = \begin{bmatrix} \lambda I_m & B \\ 0 & C \end{bmatrix}$$

Consider the characteristic polynomial of both sides.

$$\text{One has. } \det(P^{-1}AP - tI_n) = \det \begin{bmatrix} \lambda I_m - tI_m & B \\ 0 & C - tI_m \end{bmatrix}$$

$$\Leftrightarrow \det(P^t(A - tI_n)P) = \det((\lambda - t)I_m) \cdot \det(C - tI_m)$$

$$\Leftrightarrow \text{ch}_A(t) = (\lambda - t)^m \text{ch}_C(t)$$

($\text{ch}_X(t)$ denotes the characteristic polynomial of X)

Hence $m_A(\lambda) \geq m$ (since $\text{ch}_C(t)$ may have some power of $(\lambda - t)$)

Note that $m = \dim E_\lambda$, we conclude that

$$m_A(\lambda) \geq \dim E_\lambda$$

③ Trivial

□

Thm: The following are equivalent (TFAE), Given $A \in M_n(\mathbb{R})$

① A diagonalizable

② $\sum_{\text{eigenvalue } \lambda} \dim E_\lambda = n$, i.e. the geometric multiplicities of all eigenvalues is n

③ $\text{ch}_A(t) = \prod_i (x - \lambda_i)^{m_i}$ (product of linear factor)

and $m_A(\lambda) = \dim E_\lambda$ for all λ eigenvalue.

Pf: ① $\Rightarrow$ ②: A diagonalizable $\Leftrightarrow \exists P$ invertible s.t.

$$P^{-1}AP = D \text{ (diagonal)} \Leftrightarrow AP = PD$$

We know that the diagonal elements of D are precisely eigenvalues of A . (counting multiplicity). write $P = (v_1 v_2 \dots v_n)$

$$AP = (Av_1 Av_2 \dots Av_n) = P \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \\ = (\lambda_1 v_1 \quad \lambda_2 v_2 \quad \dots \quad \lambda_n v_n)$$

For example, if among $\lambda_1 \sim \lambda_n$, only $\lambda_1 = \lambda_2 = \dots = \lambda_n = \lambda$

That means $\{v_1, v_2, \dots, v_n\}$ be eigenvectors corresponding to λ

It is actually the basis of E_λ , check that!

$\Rightarrow \sum_{\lambda} \dim E_\lambda = n$ (in other word, we have enough eigenvectors to make the matrix diagonalizable.)

② $\Rightarrow$ ③ By previous theorem. $\checkmark$

③ $\Rightarrow$ ① Let $\{\lambda_1, \dots, \lambda_k\}$ be the set of distinct eigenvalues of A .
By condition given,

$$\sum_{i=1}^k \dim E_{\lambda_i} = n$$

Each E_{λ_i} has a basis β_i . We claim

$$\beta = \bigcup_{i=1}^k \beta_i \text{ is the basis of } \mathbb{R}^n$$

I leave this as an exercise for you to check

- $\text{span } \beta = \mathbb{R}^n$

- β is a linearly independent set.

proof is similar!

(Hint: Thm 1.22 of Lecture Note)

Hence, we found a basis consisting of eigenvectors $\Rightarrow$ diagonalizable $\square$

Back to the Problem!

(a) True. By previous result,

(b) Counter example: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, eigenvalue $\lambda = 1$

$$\text{Consider } E_1 = N(A - 1 \cdot I_2) = N\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$= \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ x \begin{pmatrix} 1 \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\},$$

$$\Rightarrow \dim E_1 = 1 < m_A(1) = 2 \quad \rightarrow \quad \det(A - tI) = (1-t)^2$$

By the previous result, it is not diagonalizable. //

(c) False. Counter example:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & \frac{1}{2} \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \text{diagonalizable}$$

$$AB = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \quad \text{not diagonalizable.} //$$

(4) (a) Only eigenvalue is $\lambda = 1$, $m_A(1) = 6$

$$\text{Compute } \dim E_1, \quad E_1 = N\left(\begin{pmatrix} I_3 & I_3 \\ & I_3 \end{pmatrix} - I_6\right)$$

$$= N\begin{pmatrix} 0 & I_3 \\ 0 & 0 \end{pmatrix}$$

$$\dim N\begin{pmatrix} 0 & I_3 \\ 0 & 0 \end{pmatrix} = 3 < 6 = m_A(1)$$

$\Rightarrow$ Not diagonalizable ✓

(b) Eigenvalue $\lambda_1=1$, $\lambda_2=2$, $m_a(1)=3=m_a(2)$

$$\begin{aligned}\text{Compute } \dim E_{\lambda_1} &= \dim N \left[\begin{pmatrix} I_3 & I_3 \\ 0 & 2I_3 \end{pmatrix} - I_6 \right] \\ &= \dim N \begin{bmatrix} 0 & I_3 \\ 0 & I_3 \end{bmatrix} \\ &= 3 = m_a(1)\end{aligned}$$

$$\begin{aligned}\dim E_{\lambda_2} &= \dim N \left[\begin{pmatrix} I_3 & I_3 \\ 0 & 2I_3 \end{pmatrix} - 2I_6 \right] \\ &= \dim N \begin{pmatrix} -I_3 & I_3 \\ 0 & 0 \end{pmatrix} \\ &= 3 = m_a(2)\end{aligned}$$

$\Rightarrow$ diagonalizable.

(c) Set $A = \begin{pmatrix} 3 & 0 & 0 \\ 1 & 3 & 1 \\ -1 & 0 & 2 \end{pmatrix}$

Step 1: Compute eigenvalues

$$\det(A - tI) = 0 \Rightarrow \lambda_1 = \lambda_2 = 3, \lambda_3 = 2.$$

Step 2: Compute the corresponding eigenvectors, and eigen space.

$\lambda_1 = \lambda_2 = 3$, find $v \neq 0$, s.t.

$$(A - 3I_3)v = 0 \Leftrightarrow \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{pmatrix} v = 0$$

$$\Rightarrow N(A - 3I_3) = \text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

$\lambda_3 = 2$, find $v \neq 0$, s.t.

$$(A - 2I_3)v = 0 \Leftrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & 0 \end{pmatrix} v = 0$$

$$\Rightarrow N(A - 2I_3) = \text{span} \left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$$

Step 3:

$$\lambda_1 \longleftrightarrow v_1$$

$$\lambda_2 \longleftrightarrow v_2$$

$$\lambda_3 \longleftrightarrow v_3$$

we have (by thm 1.10) $Q^{-1} A Q = D = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{pmatrix}$

where $Q = (v_1, v_2, v_3)$

Here $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{pmatrix}^{-1} A \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & & \\ & 3 & \\ & & 2 \end{pmatrix}$ //